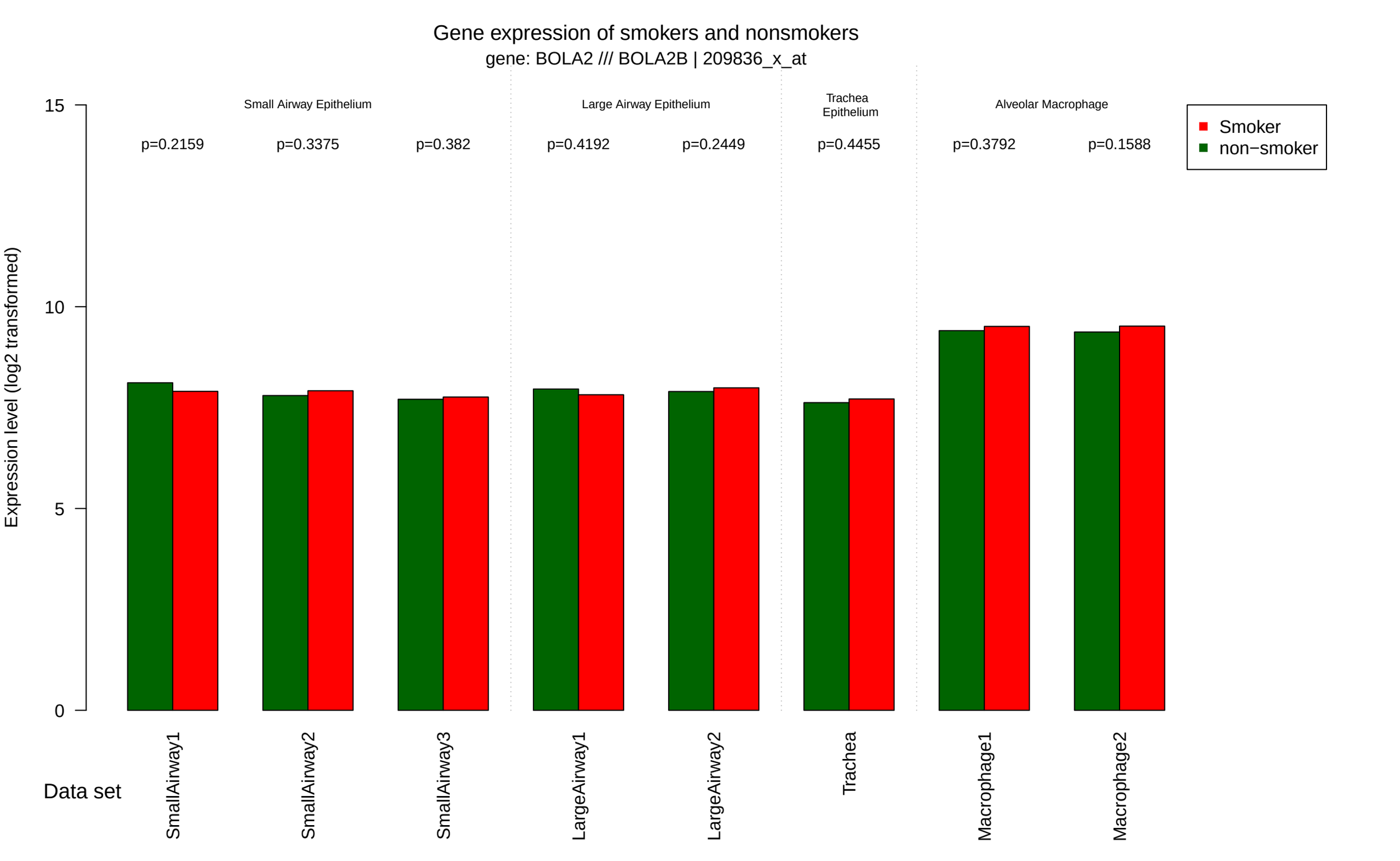
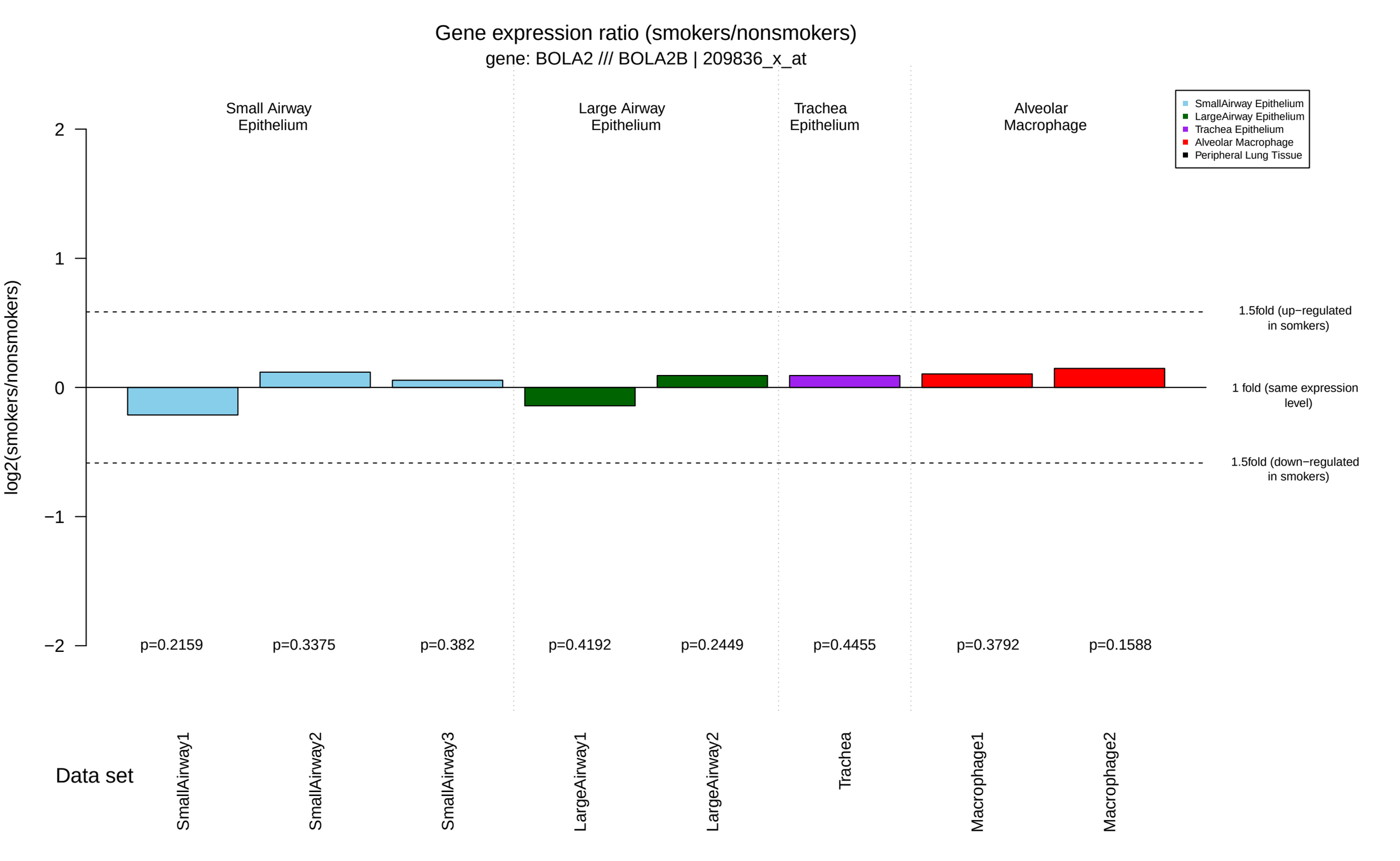
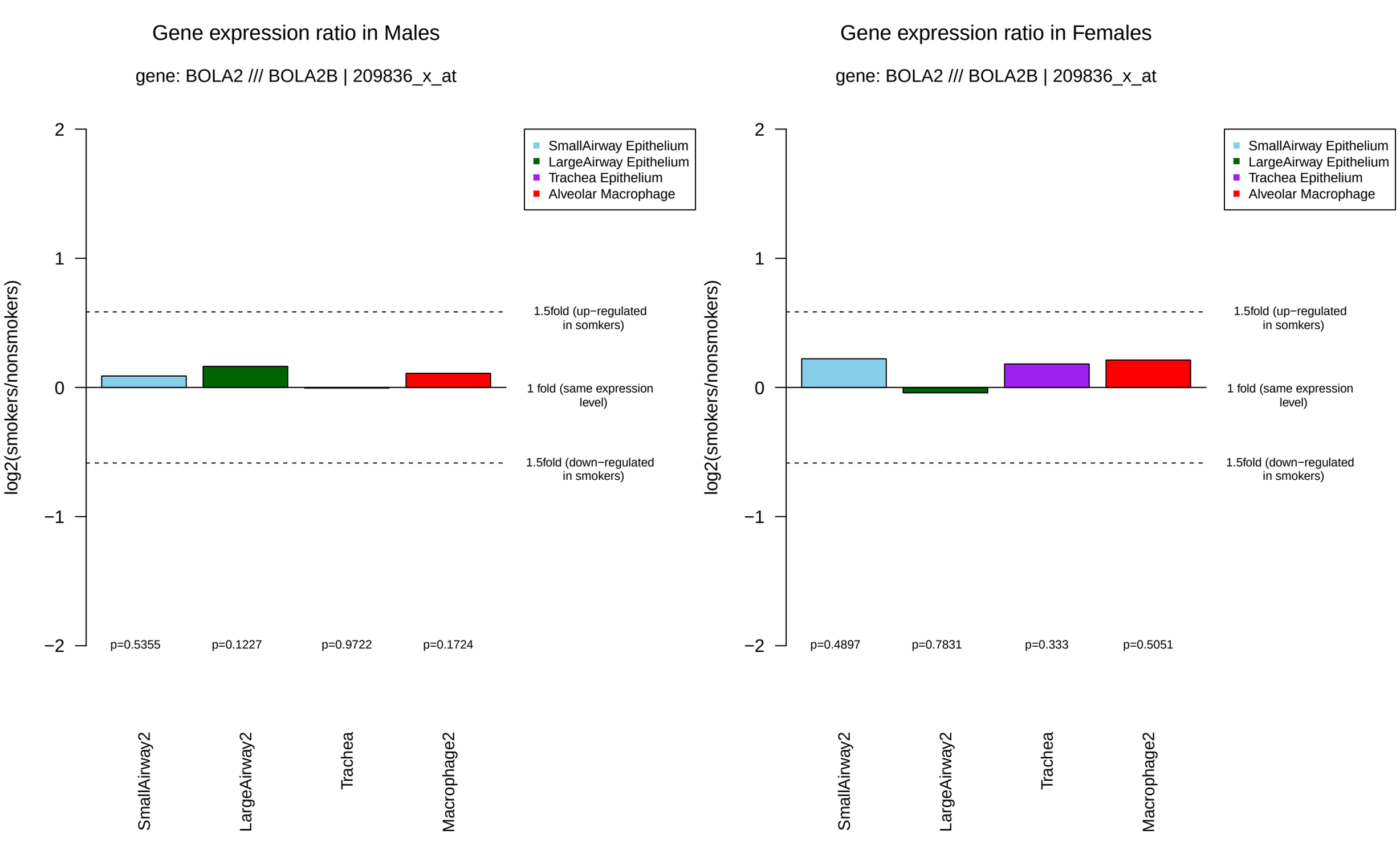
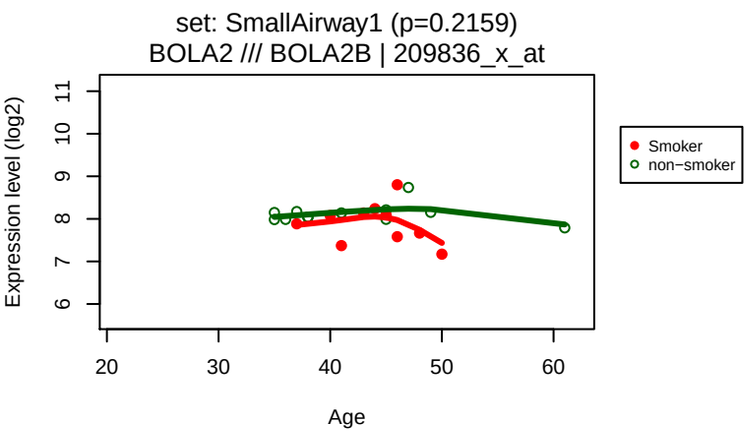
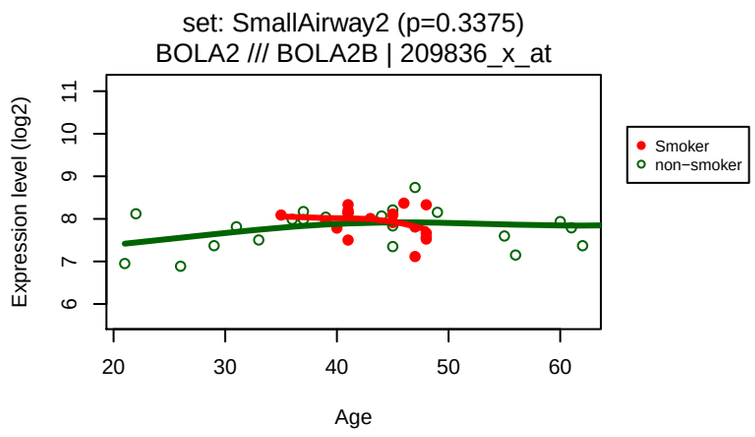
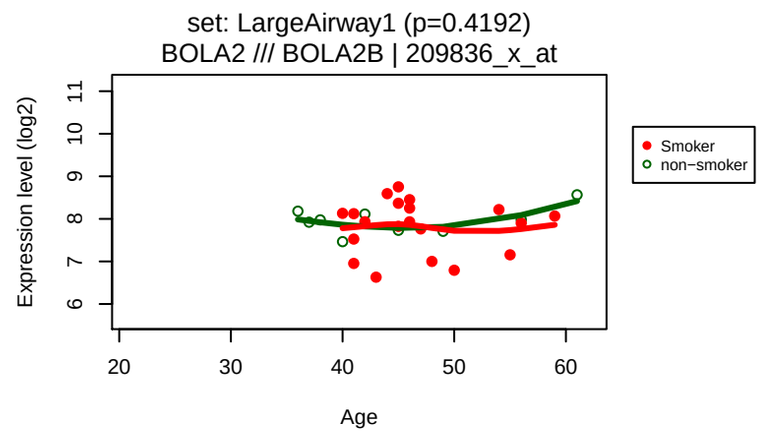

Gene expression of smokers and nonsmokers
gene: BOLA2 /// BOLA2B | 209836_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2159
p=0.3375
p=0.382
p=0.4192
p=0.2449
p=0.4455
p=0.3792
p=0.1588
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

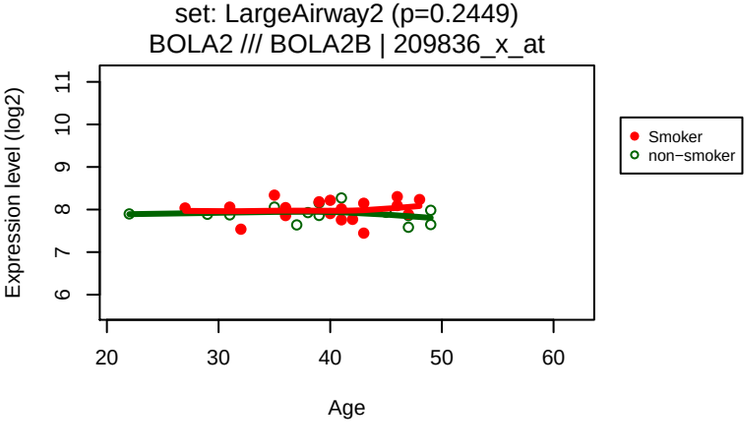

Gene expression ratio (smokers/nonsmokers)
gene: BOLA2 /// BOLA2B | 209836_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2159
p=0.3375
p=0.382
p=0.4192
p=0.2449
p=0.4455
p=0.3792
p=0.1588
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

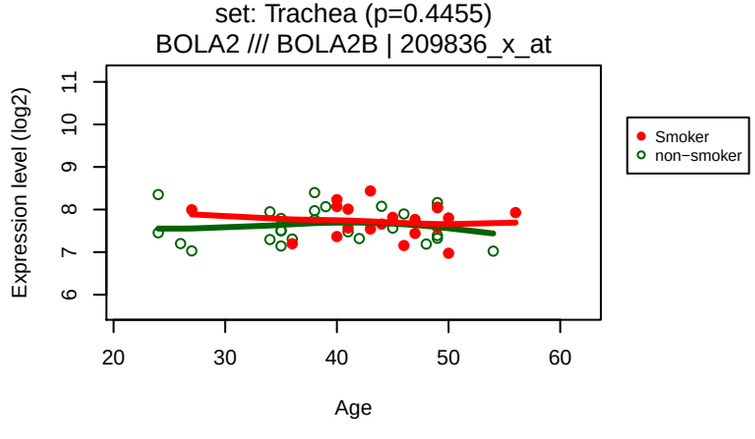

Gene expression ratio in Males
gene: BOLA2 /// BOLA2B | 209836_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5355
p=0.1227
p=0.9722
p=0.1724
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BOLA2 /// BOLA2B | 209836_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4897
p=0.7831
p=0.333
p=0.5051
SmallAirway2
LargeAirway2
Trachea
Macrophage2

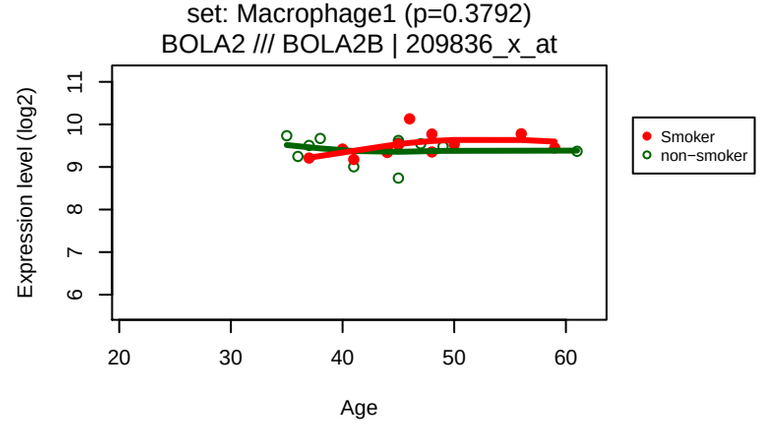

set: SmallAirway1 (p=0.2159)
BOLA2 /// BOLA2B | 209836_x_at
Expression level (log2)
Age
Smoker
non−smoker

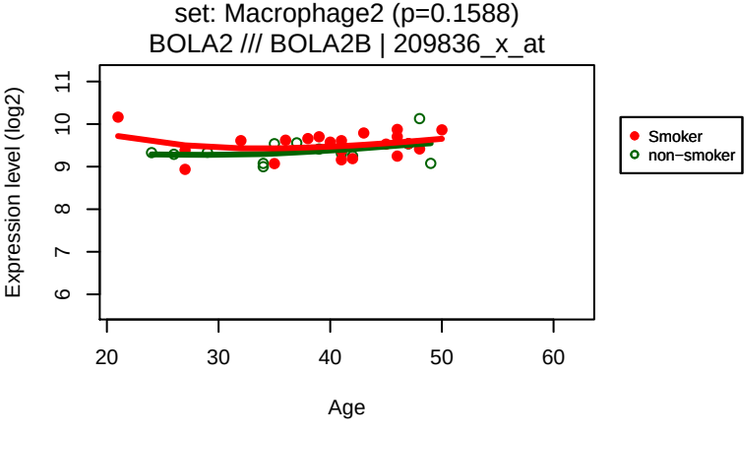

set: SmallAirway2 (p=0.3375)
BOLA2 /// BOLA2B | 209836_x_at
Expression level (log2)
Age
Smoker
non−smoker

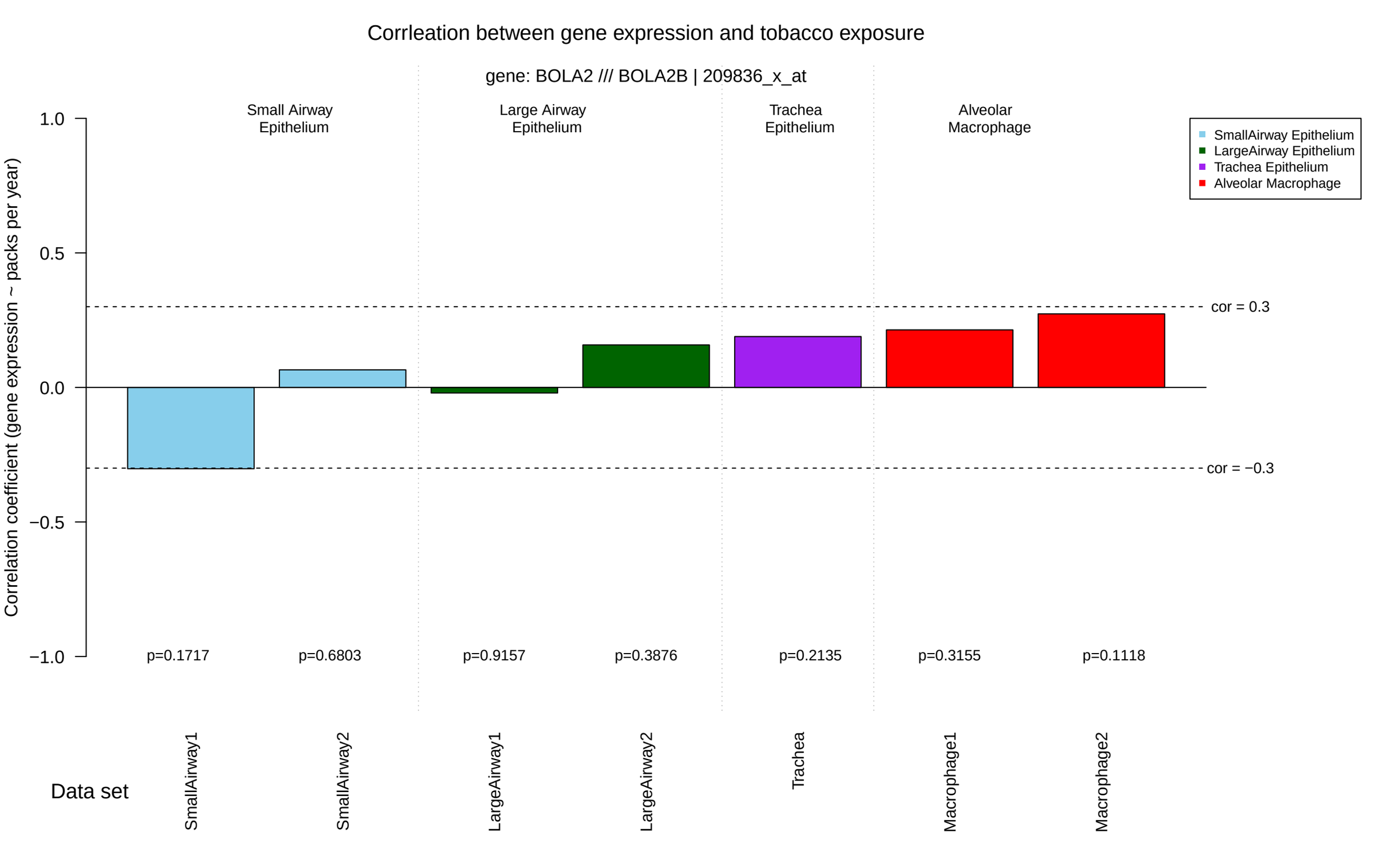

set: LargeAirway1 (p=0.4192)
BOLA2 /// BOLA2B | 209836_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2449)
BOLA2 /// BOLA2B | 209836_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4455)
BOLA2 /// BOLA2B | 209836_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3792)
BOLA2 /// BOLA2B | 209836_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1588)
BOLA2 /// BOLA2B | 209836_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BOLA2 /// BOLA2B | 209836_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.1717
p=0.6803
p=0.9157
p=0.3876
p=0.2135
p=0.3155
p=0.1118
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2